

EXP NO: 1	SETTING UP THE ENVIRONMENT AND PREPROCESSING THE DATA
Date:16-7-26	

AIM:

To set up a fully functional machine learning development environment and to perform data preprocessing operations like handling missing values, encoding categorical variables, feature scaling, and splitting datasets.

ALGORITHM:

1. Install Required Libraries:

- Install numpy, pandas, matplotlib, seaborn, and scikit-learn using pip.

2. Import Libraries.

3. Load Dataset:

- Load any dataset (e.g., Netflix or Iris) using pandas.

4. Data Exploration:

- Use `df.info()`, `df.describe()`, `df.isnull().sum()` to understand the data.

5. Handle Missing Values:

- Use `.fillna()` or `.dropna()` depending on the strategy.

6. Encode Categorical Data:

- Use `pd.get_dummies()` or `LabelEncoder`.

7. Feature Scaling:

- Normalize or standardize the numerical features using `StandardScaler` or

`MinMaxScaler`.

8. Split Dataset:

- Use `train_test_split()` from `sklearn` to create training and testing sets.

9. Display the Preprocessed Data.

CODE:

```
In [12]: import pandas as pd
import numpy as np
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.model_selection import train_test_split

df = pd.read_csv("netflix_titles.csv")
print("First 5 Rows")
print(df.head())
print("\nDataset Information")
print(df.info())
print("\nStatistical Summary")
print(df.describe(include='all'))
print("\nMissing Values")
print(df.isnull().sum())
categorical_columns = ['director', 'cast', 'country',
                       'rating', 'duration']

for col in categorical_columns:
    df[col].fillna("Unknown", inplace=True)
df['date_added'].fillna(df['date_added'].mode()[0], inplace=True)
df.dropna(subset=['title'], inplace=True)
le = LabelEncoder()
columns_to_encode = [
    'type',
    'title',
    'director',
    'cast',
    'country',
    'rating',
    'duration',
    'listed_in'
]

for col in columns_to_encode:
    df[col] = le.fit_transform(df[col].astype(str))

scaler = StandardScaler()

numeric_columns = ['release_year']

df[numeric_columns] = scaler.fit_transform(df[numeric_columns])

X = df[['type', 'release_year', 'rating',
        'duration', 'listed_in']]

y = df['type']

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

print("\nPreprocessed Dataset")
print(df.head())

print("\nTraining Data Shape:", X_train.shape)
print("Testing Data Shape:", X_test.shape)
```

OUTPUT:

```
First 5 Rows
  show_id  type      title      director \
0      s1  Movie  Dick Johnson Is Dead  Kirsten Johnson
1      s2  TV Show      Blood & Water      NaN
2      s3  TV Show      Ganglands      Julien Leclercq
3      s4  TV Show  Jailbirds New Orleans      NaN
4      s5  TV Show      Kota Factory      NaN

      cast      country \
0      NaN  United States
1  Ama Qamata, Khosi Ngema, Gail Mablane, Thaban...  South Africa
2  Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...      NaN
3      NaN      NaN
4  Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...  India

      date_added  release_year  rating  duration \
0  September 25, 2021      2020  PG-13      90 min
1  September 24, 2021      2021  TV-MA  2 Seasons
2  September 24, 2021      2021  TV-MA  1 Season
3  September 24, 2021      2021  TV-MA  1 Season
4  September 24, 2021      2021  TV-MA  2 Seasons

      listed_in \
0      Documentaries
1  International TV Shows, TV Dramas, TV Mysteries
2  Crime TV Shows, International TV Shows, TV Act...
3      Docuseries, Reality TV
4  International TV Shows, Romantic TV Shows, TV ...

      description
0  As her father nears the end of his life, filmm...
1  After crossing paths at a party, a Cape Town t...
2  To protect his family from a powerful drug lor...
3  Feuds, flirtations and toilet talk go down amo...
4  In a city of coaching centers known to train I...

Dataset Information
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   show_id     8807 non-null   object
1   type        8807 non-null   object
2   title       8807 non-null   object
3   director    6173 non-null   object
4   cast        7982 non-null   object
5   country     7976 non-null   object
6   date_added  8797 non-null   object
7   release_year 8807 non-null   int64
8   rating      8803 non-null   object
9   duration    8804 non-null   object
10  listed_in   8807 non-null   object
11  description  8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
None
```

Statistical Summary

	show_id	type	title	director	\
count	8807	8807	8807	6173	
unique	8807	2	8807	4528	
top	s1	Movie	Dick Johnson Is Dead	Rajiv Chilaka	
freq	1	6131	1	19	
mean	NaN	NaN	NaN	NaN	
std	NaN	NaN	NaN	NaN	
min	NaN	NaN	NaN	NaN	
25%	NaN	NaN	NaN	NaN	
50%	NaN	NaN	NaN	NaN	
75%	NaN	NaN	NaN	NaN	
max	NaN	NaN	NaN	NaN	

	cast	country	date_added	release_year	\
count	7982	7976	8797	8807.000000	
unique	7692	748	1767	NaN	
top	David Attenborough	United States	January 1, 2020	NaN	
freq	19	2818	109	NaN	
mean	NaN	NaN	NaN	2014.180198	
std	NaN	NaN	NaN	8.819312	
min	NaN	NaN	NaN	1925.000000	
25%	NaN	NaN	NaN	2013.000000	
50%	NaN	NaN	NaN	2017.000000	
75%	NaN	NaN	NaN	2019.000000	
max	NaN	NaN	NaN	2021.000000	

	rating	duration	listed_in	\
count	8803	8804	8807	
unique	17	220	514	
top	TV-MA	1 Season	Dramas, International Movies	
freq	3207	1793	362	
mean	NaN	NaN	NaN	
std	NaN	NaN	NaN	
min	NaN	NaN	NaN	
25%	NaN	NaN	NaN	
50%	NaN	NaN	NaN	
75%	NaN	NaN	NaN	
max	NaN	NaN	NaN	

	description
count	8807
unique	8775
top	Paranormal activity at a lush, abandoned prope...
freq	4
mean	NaN
std	NaN
min	NaN
25%	NaN
50%	NaN
75%	NaN
max	NaN

Missing Values

```
show_id      0
type         0
title        0
director     2634
cast         825
country      831
date_added   10
release_year  0
rating       4
duration     3
listed_in    0
description  0
dtype: int64
```

Preprocessed Dataset

```
show_id  type  title  director  cast  country  date_added \
0      s1     0   1975     2295  7246     603  September 25, 2021
1      s2     1  1091     4303   409     426  September 24, 2021
2      s3     1  2651     2105  6296     736  September 24, 2021
3      s4     1  3506     4303  7246     736  September 24, 2021
4      s5     1  3861     4303  4815     251  September 24, 2021
```

```
release_year  rating  duration  listed_in \
0      0.659930      7       210       274
1      0.773324     11       110       414
2      0.773324     11        0       242
3      0.773324     11        0       297
4      0.773324     11       110       393
```

description

```
0 As her father nears the end of his life, filmm...
1 After crossing paths at a party, a Cape Town t...
2 To protect his family from a powerful drug lor...
3 Feuds, flirtations and toilet talk go down amo...
4 In a city of coaching centers known to train I...
```

Training Data Shape: (7045, 5)

Testing Data Shape: (1762, 5)

RESULT:

The Python environment was successfully set up and the dataset was pre-processed by handling missing values, encoding categorical data, performing feature scaling, and splitting the data into training and testing sets. The dataset is now ready for model training and analysis.